



OBJECT ORIENTED SOFTWARE ENGINEERING

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4. Functional Models – Activity Diagram

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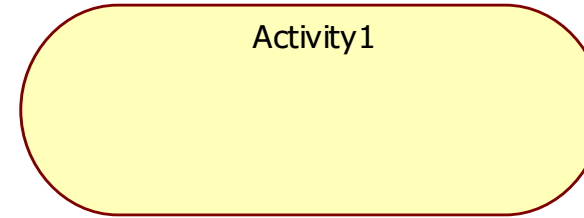
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- Activity models are expressed through **Activity diagram**.
- An activity diagram shows **the sequence of steps that make up a complex process, such as algorithm or work flow**.
- It shows **flow of control** similar to sequence diagrams, but **focuses on operations rather than on objects**.
- Activity diagrams are most useful during the **early stages of designing algorithms and work flow**.

Components used to construct activity diagrams are:

- **Elongated oval** - shows activities and
- **arrows** - show their sequencing.
- **Diamond** - shows a decision point and
- **heavy bar** - shows splitting or merging of concurrent thread.

1. **Activities** - represented as elongated oval



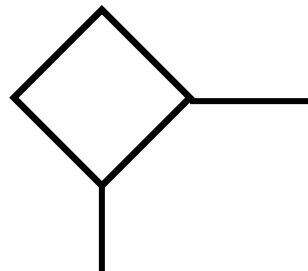
The steps of an activity diagram are **operations**, specifically **activities from the state model**.

Completion of activity is a **completion of event** and usually indicate that the next activity can be started.

Unlabelled arrow $\longrightarrow$ shows first activity must complete before the second activity can began.

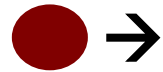
2. Branch:

- If there is more than **one successor to an activity**, each arrow maybe labeled with a **condition in square** bracket.eg:- [failure] [success].
- All **subsequent conditions are tested** when an activity completes. If one condition is satisfied, its arrow indicates next activity to perform.
- As a notational convenience, a **diamond shows a branch into multiple successors**.



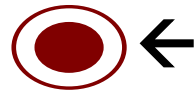
3. Initiation:

A **solid circle with an out going arrow** shows the **starting point** of an activity diagram.



4. Termination:

A **bulls eye** (a solid circle surrounded by a hollow circle) shows the **termination point**, this symbol has only **incoming arrow**.



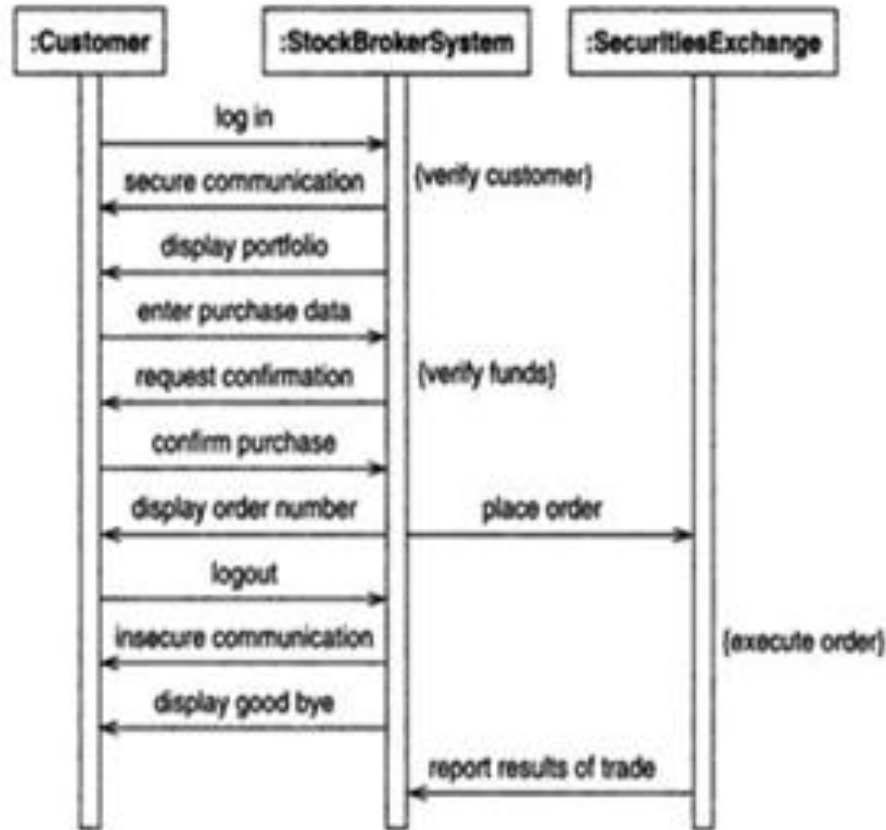


Figure 7.5 Sequence diagram for a session with an online stock broker. A sequence diagram shows the participants in an interaction and the sequence of messages among them.

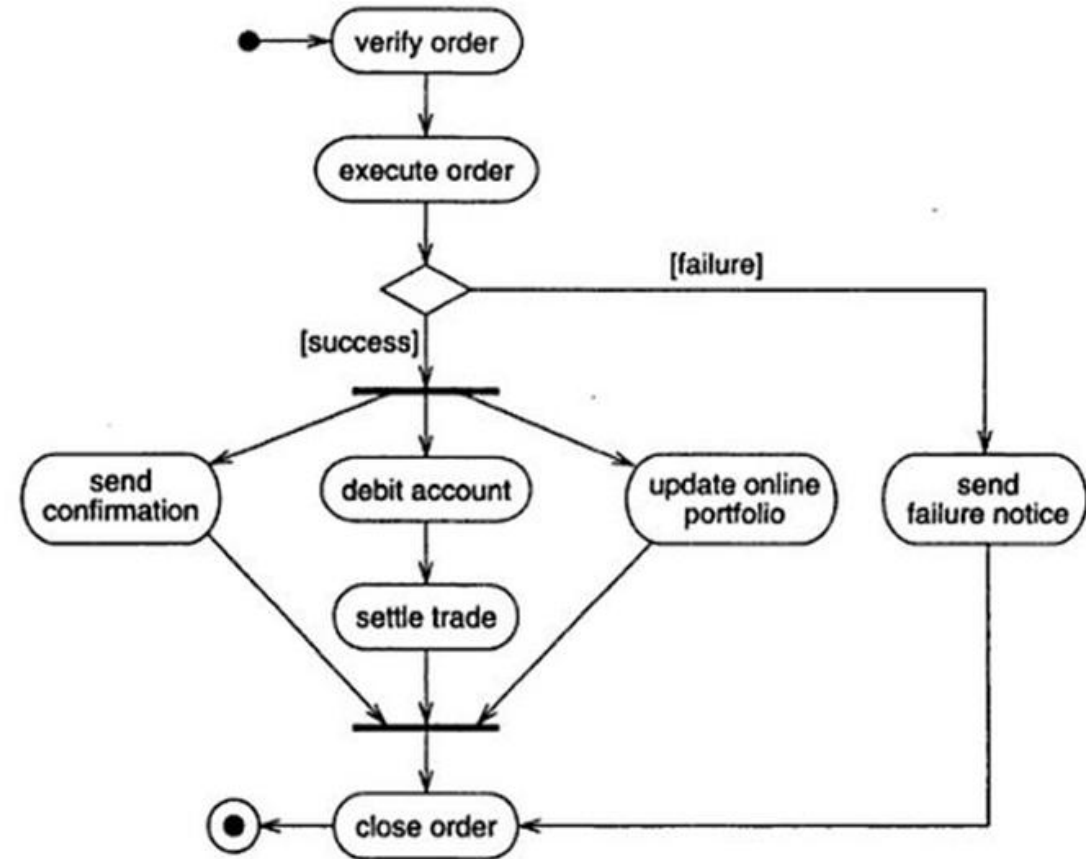


Figure Activity diagram for stock trade processing. An activity diagram shows the sequence of steps that make up a complex process.

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Activity diagram example

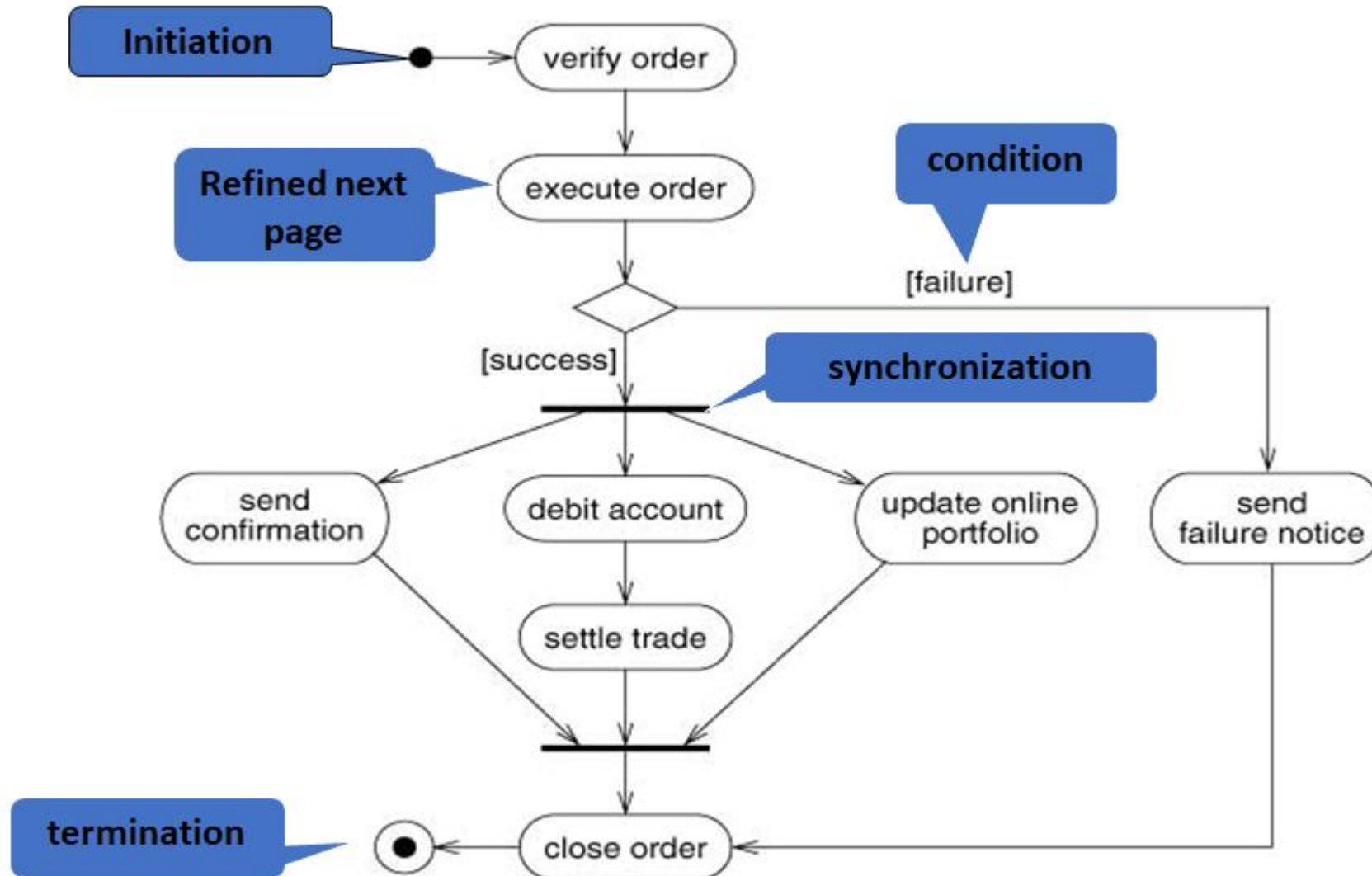


Figure 7.9 Activity diagram for stock trade processing. An activity diagram shows the sequence of steps

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Guidelines for Activity diagram

1. **Don't misuse activity diagram**:-It should **not** be used as an **excuse** to develop software via **flow chart**.
2. **Level diagrams**:-Activities on a diagram should be at a **consistent level of detail**.
3. **Be careful with branches and conditions** :-If there are conditions, at least one must be satisfied when an activity completes, conditions using an else condition.



THANK YOU

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